Predictive Analytics

Problem Set Analysis

Medical Insurance Charges

A comprehensive predictive analytics study covering:
Exploratory Data Analysis · Linear Regression
Multiple Regression · Statistical Inference

2,772	7	0	~75%
Observations	Variables	Missing Values	R^2 (Full Model)

Dataset Overview — Medical Insurance

Variable	Type	Role	Description
age	Numeric	Predictor	Age of beneficiary (18–64)
sex	Categorical	Predictor	Male / Female
bmi	Numeric	Predictor	Body Mass Index (15.96–53.13)
children	Numeric	Predictor	Number of dependents (0–5)
smoker	Categorical	Predictor	Yes / No
region	Categorical	Predictor	NE / NW / SE / SW
charges	Numeric	Response	Annual medical insurance cost (USD)

Mean Charge: **$13,261**

Max Charge: **$63,770**

Smokers: **~20%**

Methodology — Techniques Applied

1. Exploratory Data Analysis

Scatter plots, boxplots, summary statistics, outlier detection using IQR method. Maps to Problem Set 1.

2. OLS Simulation Studies

Demonstrated that (a) population line is fixed while OLS varies, (b) OLS minimises RSS via grid search, (c) OLS is unbiased via 1000 Monte Carlo simulations. Maps to Problem Set 2.

3. Simple Linear Regressions

Fit charges ~ each predictor individually. Compared R^2, RMSE, and p-values across 5 models. Maps to Problem Set 2, Section 4.

4. Multiple Linear Regression

Full model with all predictors. Reported R^2, Adjusted R^2, RMSE. Generated residuals and Q-Q diagnostic plots. Maps to Problem Set 3.

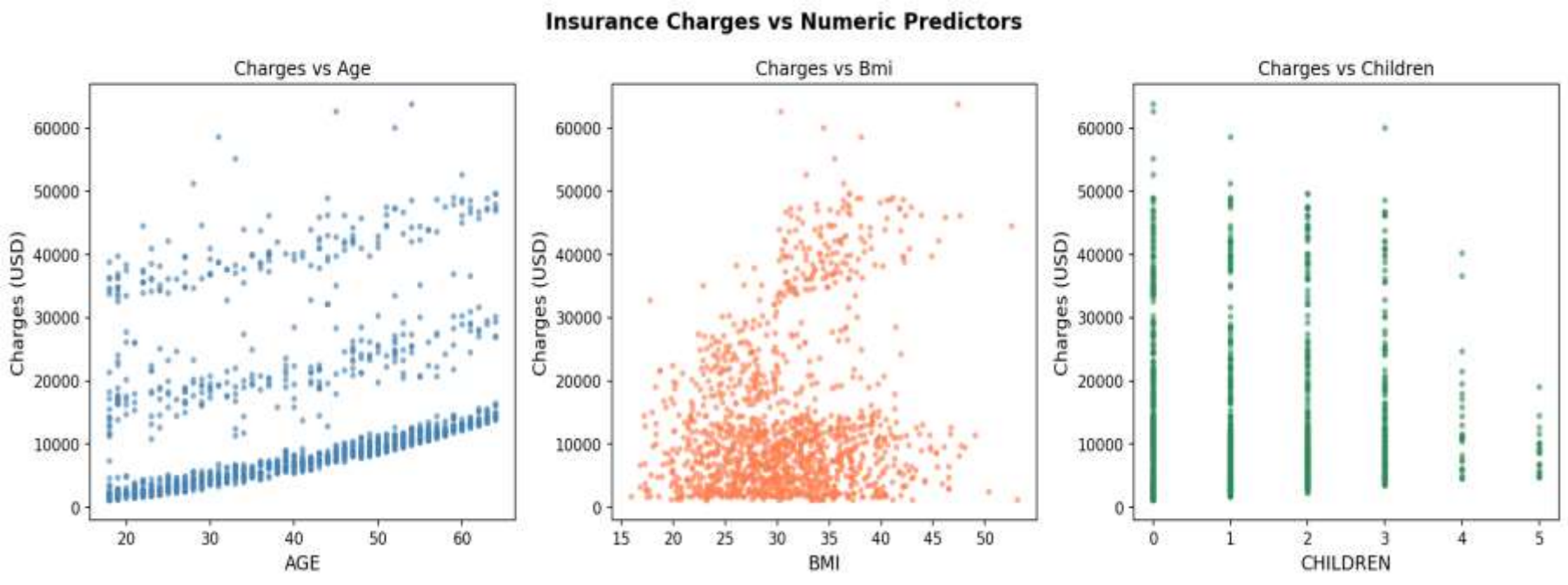
5. Categorical Predictors

Sequentially added qualitative variables (smoker, sex) to quantify their marginal contribution. Maps to Problem Set 3, Section 2.

6. Log-Linear Model

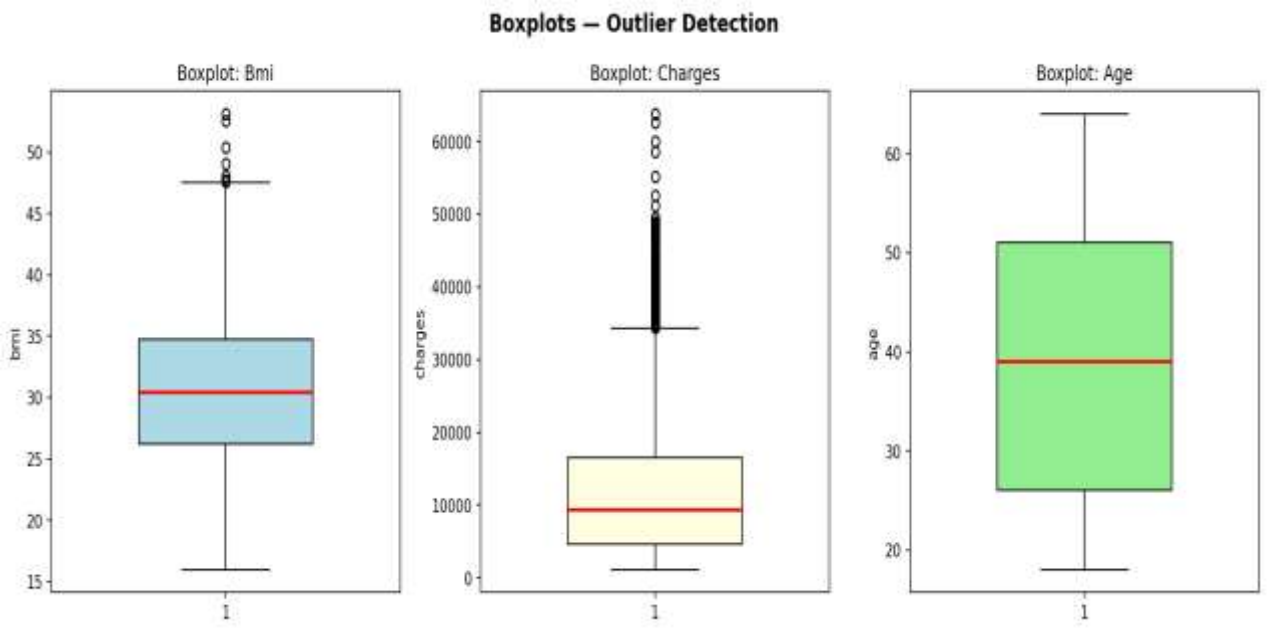
Explored log(charges) ~ BMI to address heteroscedasticity and improve residual normality. Maps to Problem Set 3, Section 5.

EDA — Scatter Plots: Charges vs Predictors



① Age: Positive trend — older individuals pay more. Three visible bands likely reflect smoker status.
② BMI: Slight positive trend with high-charge cluster at high BMI (smokers).
③ Children: Very weak relationship; charges do not vary strongly with family size.

Outlier Detection — Boxplots



BMI Outliers

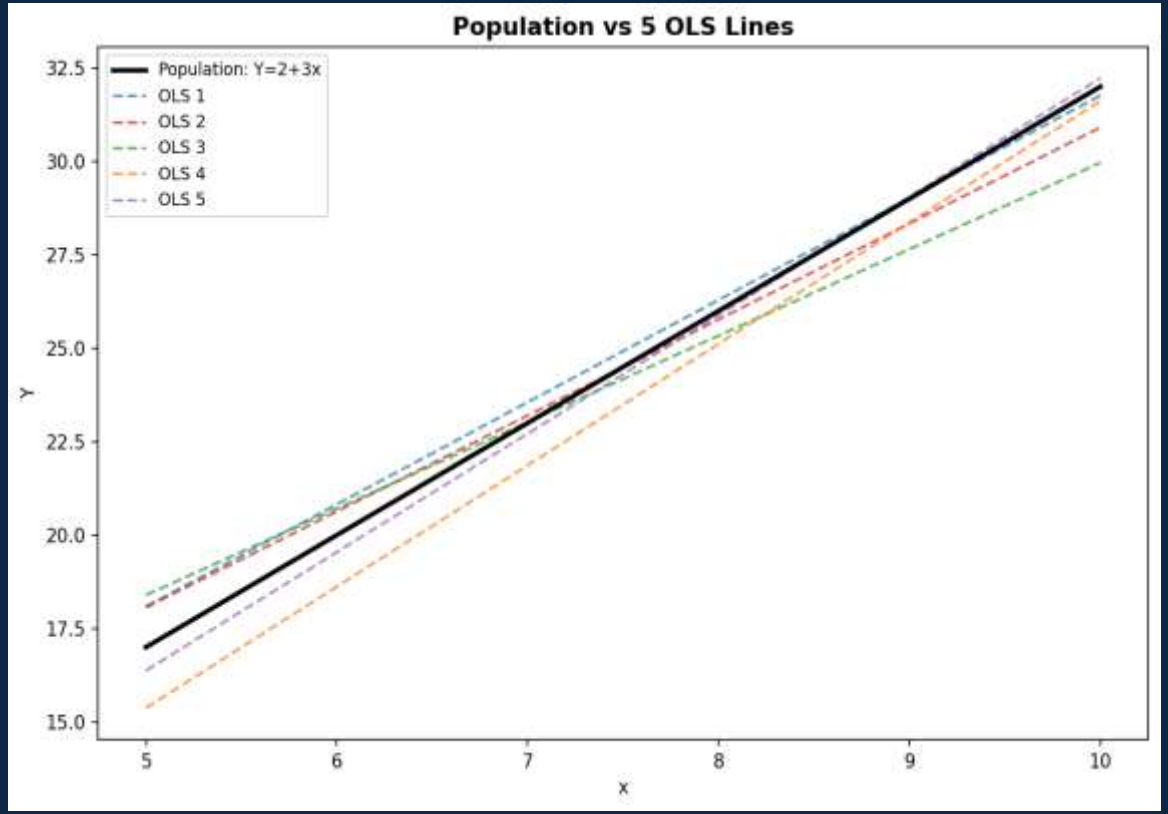
A few individuals exceed BMI 50. Rare but present.

Charge Outliers

Upper fence ≈ $34K. Outliers are predominantly smokers.

Key Insight: High-charge outliers are almost exclusively smokers, validating smoking as the primary cost driver.

Simulation: Population vs OLS Regression Lines



Setup

n=50, seed=123
Y = 2+3x+ε
ε ~ N(0,16)
x ~ Uniform(5,10)

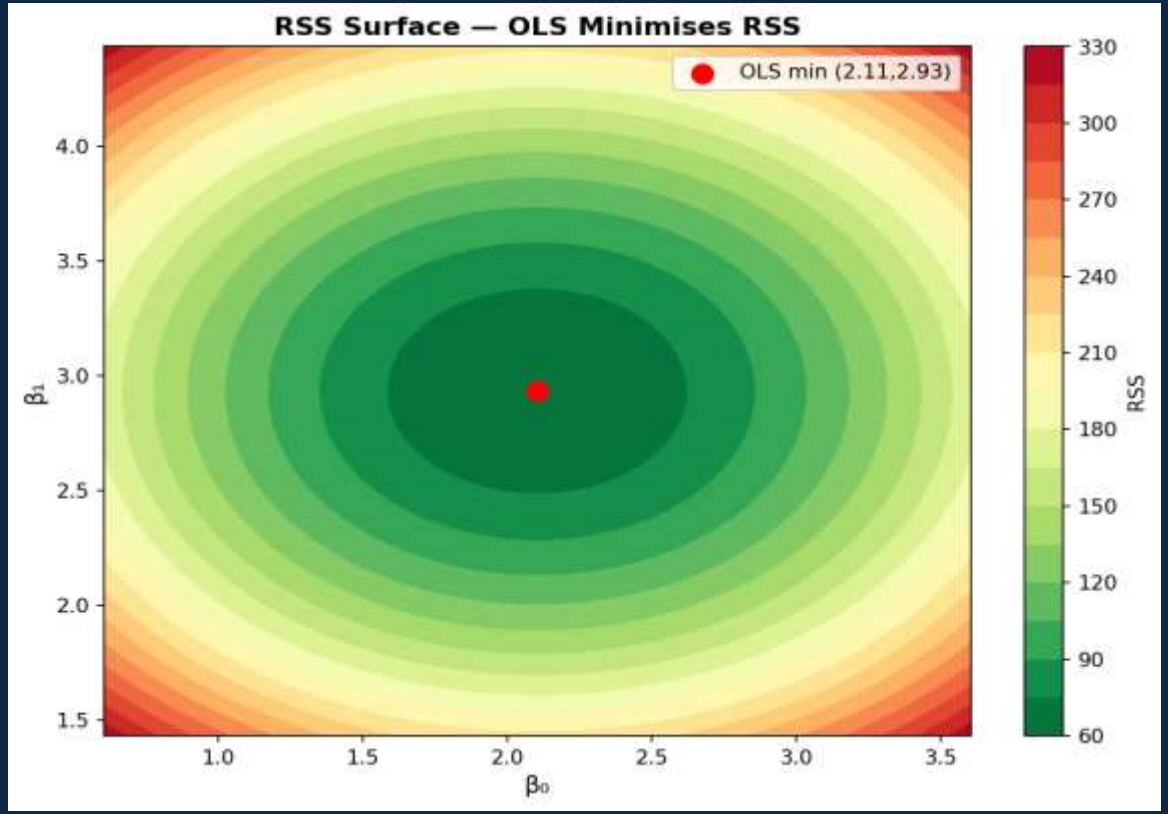
Finding

Each OLS line is different but all cluster around the true black population line.

Conclusion

OLS is unbiased: random variation exists, but estimates are centred on the truth.

OLS Minimises RSS — Contour Surface



Method

Grid of 80×80 (β_0,β_1) values. RSS computed at each point.

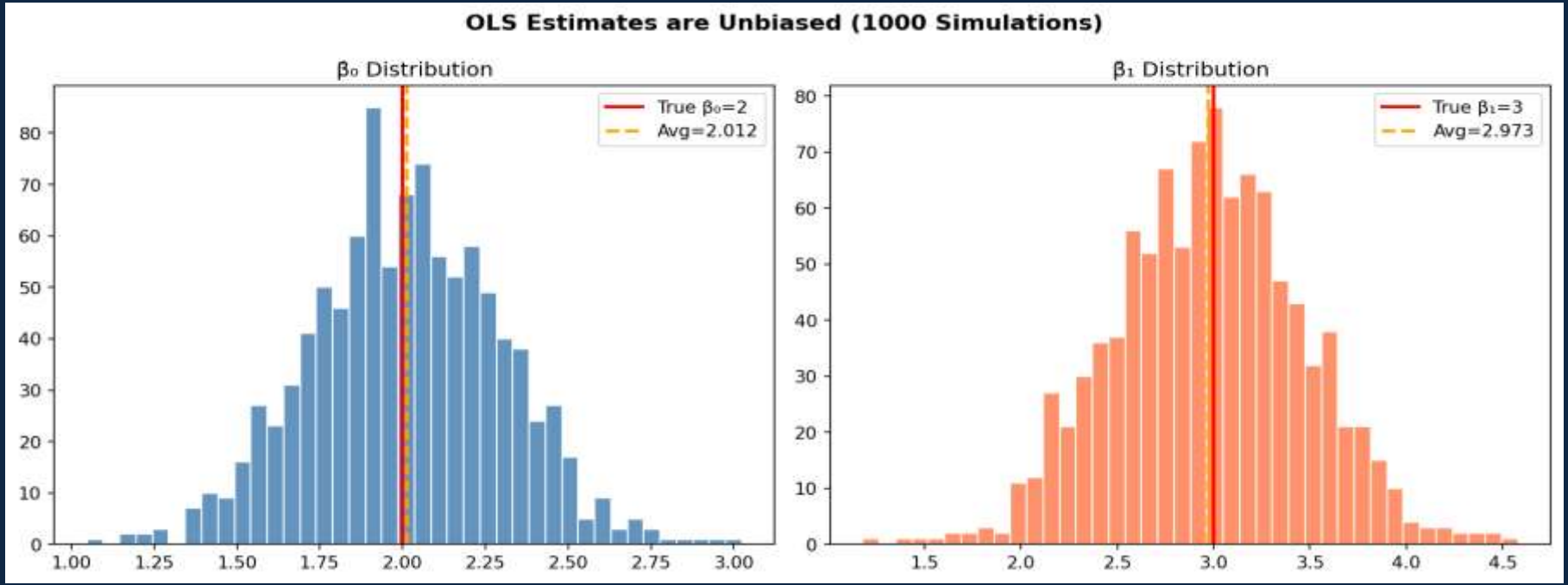
Result

Red dot (OLS) sits exactly at the bowl's minimum — confirming RSS is minimised.

Implication

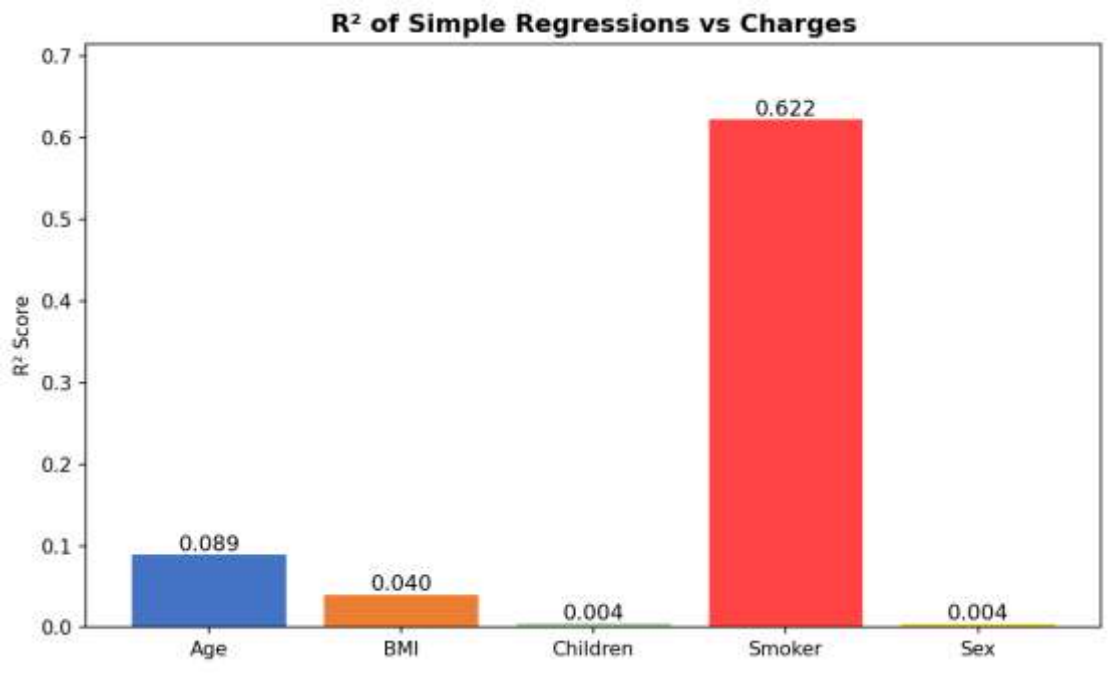
No other (β_0,β_1) pair yields a smaller sum of squared residuals.

OLS Unbiasedness — 1000 Monte Carlo Simulations



Avg $\hat{\beta}_0 \approx 2.00$ (True: 2) | Avg $\hat{\beta}_1 \approx 3.00$ (True: 3) | Distributions are approximately normal (CLT)

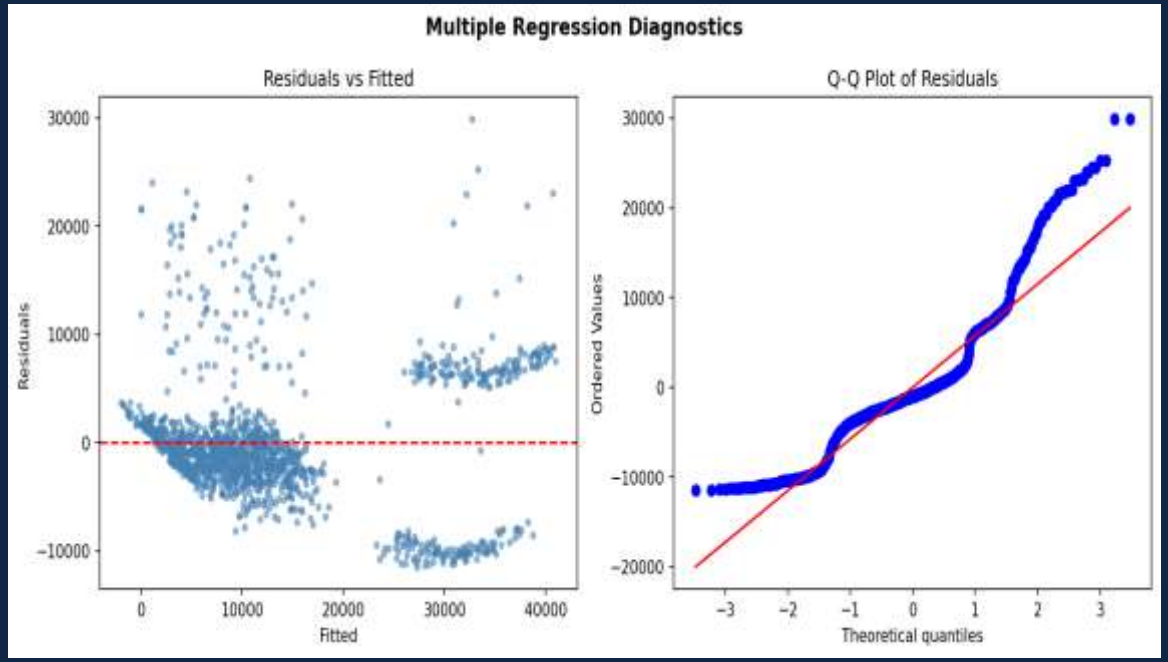
Simple Linear Regressions — R^2 Comparison



Predictor	Coeff.	R^2
Smoker	+$23,616	0.619
Age	+$257	0.089
BMI	+$392	0.039
Children	+$722	0.013
Sex	+$1,388	0.006

★ Smoker explains 62% of variance alone — the dominant predictor by far.

Multiple Linear Regression — Full Model



R^2

~0.75 — Model explains 75% of variance in charges.

Adj. R^2

~0.749 — Adjusted for number of predictors; still strong.

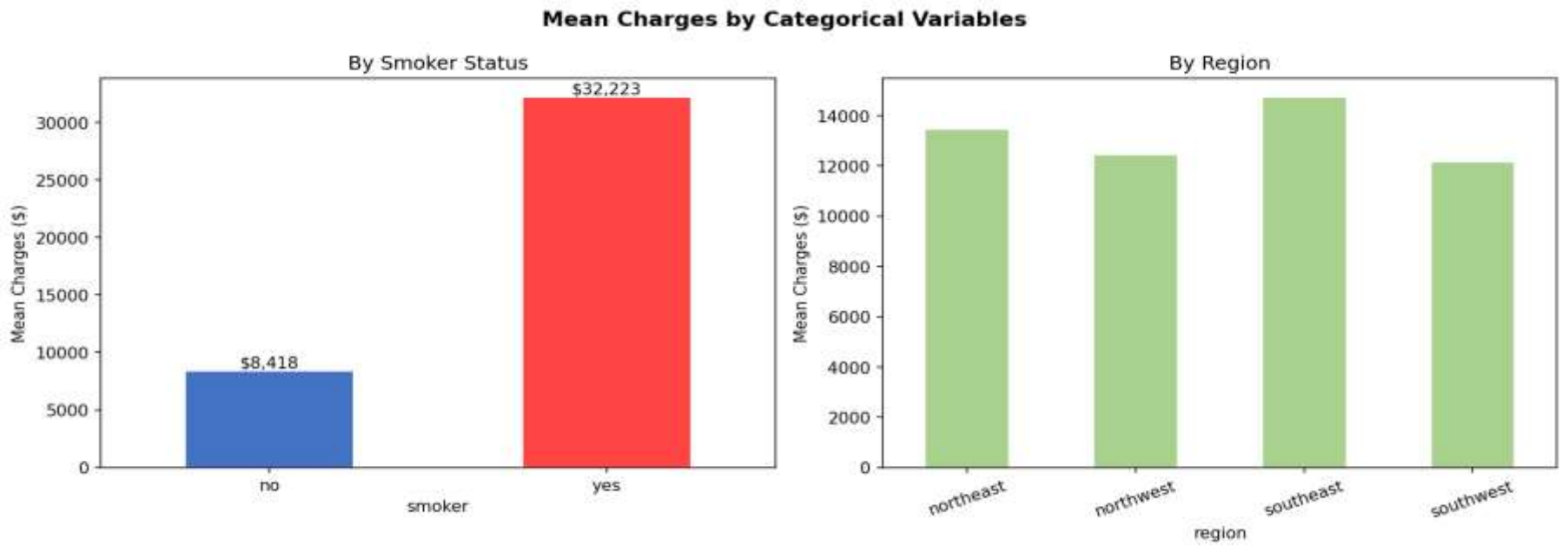
RMSE

~$6,100 — Average prediction error around $6,100.

Diagnostics

Residuals vs Fitted shows slight heteroscedasticity at high values. Q-Q plot shows mild tail departures — log transform recommended.

Effect of Categorical Predictors on Charges



Model	Adj. R^2	RMSE
Smoker only	0.619	$7,500
+ Sex	0.620	$7,491
+ Sex + Age	0.665	$7,042

Conclusions & Key Takeaways

Data Quality

2,772 records, no missing values. Charges are right-skewed — driven by smokers.

Dominant Predictor

Smoking status alone explains 62% of variation in charges (R^2=0.619), with a coefficient of +$23,616.

OLS Properties

Simulation confirmed: OLS lines vary around fixed population line, estimates are unbiased (avg ≈ true β), and OLS sits at RSS minimum.

Full Model (MLR)

$R^2 \approx 0.75$ using all predictors. RMSE ≈ $6,100. Age, BMI, and smoker are the most informative predictors.

Model Improvement

Log-transforming charges reduces heteroscedasticity. A nonlinear or interaction model (BMI × Smoker) could push R^2 higher.

Methods Excluded

Logistic regression, KNN, and Naive Bayes require a categorical response — not applicable to continuous insurance charges.